FIT3155: Week 8 Lab Sheet

(Questions are based on Week 7 topic dealing with Fibonacci heaps.)

Objectives: Develop a working understanding of the structure and operations of Fibonacci heaps, leading to how they support efficient amortized time for priority queue operations.

Conceptual exercises

1. You studied Dijkstra's algorithm in FIT2004. Suppose you are given a general heap data structure that supports the operations `extract-min` and `decrease-key`. Express the overall time complexity of Dijkstra's algorithm on a graph $G(V, E, W)$ in terms of:

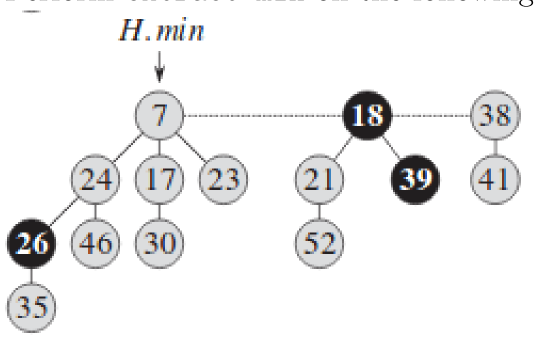
   - the number of vertices $|V|$,
   - the number of edges $|E|$, and
   - the time complexities of `extract-min` and `decrease-key` denoted by $T_{\texttt{extract-min}}$ and $T_{\texttt{decrease-key}}$.

2. `Insert` the following elements into a Fibonacci heap:

$$3, 8, 10, 11, 1, 12, 2, 17$$

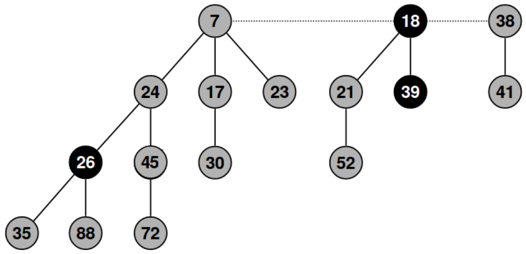
   and then perform `extract-min` on the resulting heap.

3. Perform `extract-min` on the following Fibonacci heap:



4. What are the worst-case time complexities of `insert` and `extract-min` operations?

5. Starting from the following state of a Fibonacci heap:



Run the following sequence of operations, and after each step, draw the resultant heap:

- ☐ (a) `decrease-key` of 45 to 40.
- ☐ (b) `decrease-key` of 40 to 12.
- ☐ (c) `decrease-key` of 35 to 1.
- ☐ (d) `extract-min`.

☐ 6. Let F_n denote the n-th Fibonacci number, where $F_n = F_{n-1} + F_{n-2}$. Prove using induction that $\sum_{i=0}^{n} F_i = F_{n+2} - 1$.

☐ 7. Using induction, show that for all integers $n \geq 0$, the $(n+2)$-th Fibonacci number satisfies $F_{n+2} \geq \phi^n$, where $\phi = \frac{1+\sqrt{5}}{2}$ (aka, the Golden Ratio).

☐ 8. Prove the maximum degree bound of $O(\log N)$ on any node in a Fibonacci Heap containing N elements.

Implementation exercise

(A message of caution: 'Vibe coding' with GenAI will hinder your learning of algorithms and data structures. The degree (and this unit) you have enrolled into is a specialist degree (unit). By definition it expects one to demonstrate specialist understanding of topics. This understanding comes from making a concerted effort, via thinking, implementing, debugging etc.)

☐ 1. Implement a Fibonacci heap data structure supporting the following operations:

- `insert`
- `min`
- `extract-min`
- `merge`
- `decrease-key`
- `delete`

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END
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